Quantification: 548_hyp_HP_z08c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	15
interior lacuna count	13
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 13)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3251.69	3031	2509.91	px^2
major_axis_length_px	lacuna shape	86.1545	87.8405	42.4335	px
minor_axis_length_px	lacuna shape	47.1287	47.2353	20.5564	px
aspect_ratio	lacuna shape	1.8234	1.6331	0.496	unitless
eccentricity	lacuna shape	0.793	0.7906	0.1084	unitless
solidity	lacuna shape	0.7971	0.8013	0.1029	unitless
roots_count	headline	10.7692	8	6.7101	count
ring_length_r30_px	headline	336.692	348	177.043	px
ring_length_r60_px	reported	640.923	654	283.464	px
owned_length_px	ownership-dependent	933.865	952.051	598.07	px
edge_count	ownership-dependent	46.1538	49	29.1772	count
mean_edge_length_px	ownership-dependent	19.8455	19.7353	3.6103	px
perimeter_px	normalised	288.307	274.007	151.225	px
ring_area_r30_px2	normalised	9624.92	10180	4013.21	px^2
ring_area_r60_px2	normalised	21102.8	22542	7400.87	px^2
in_frame_fraction_r30	normalised	0.9809	1	0.0454	unitless
in_frame_fraction_r60	normalised	0.9558	1	0.0915	unitless
ring_density_r30	normalised	0.0347237	0.0366405	0.00977197	px^-1
ring_density_r60	normalised	0.0302466	0.030294	0.00607405	px^-1
roots_per_100px_perimeter	normalised	3.8147	3.8059	1.3723	per 100 px
ring_attached_length_r30_px	network v2	307.308	336	175.214	px
ring_attached_length_r60_px	network v2	472.538	560	265.839	px
ring_length_w_r30_px	network v2	384.866	396.049	202.287	px
ring_length_w_r60_px	network v2	731.929	741.713	321.17	px
sholl_crossings_r10	network v2	8.8462	8	4.4318	count
sholl_crossings_r20	network v2	8.6154	8	4.9923	count
sholl_crossings_r30	network v2	7.7692	6	5.1015	count
loop_count	loops, report only	2.1538	2	1.864	count
loop_edge_share	loops, report only	0.1384	0.0972	0.1207	share
ring_width_mean_r30_px	width	7.3024	7.6251	0.9582	px

Per field
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measure	value	units
analysed_area_px2	996517	px^2
total_skeleton_length_px	25179	px
canalicular_length_density_per_px	0.025267	px^-1
junction_count	494	count
junction_density_per_px2	0.00049573	px^-2
lacuna_count	15	count

lacunae_per_px2	1.505e-05	px^-2
skeleton_component_count	372	count
mean_component_length_px	67.6855	px
median_component_length_px	21	px
field_length_density_w_per_px	0.0288708	px^-1
field_density_without_flagged_per_px	0.025267	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	266	px
unattached_component_count	15	count
canalicular_length_density_attached_per_px	0.025267	px^-1
field_length_density_with_unattached_per_px	0.0255339	px^-1
loop_count	37	count
loop_edge_share	0.0956	share
edge_count	1339	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.6204	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	25179	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1925	65.1309	45.5827	1.4289	0.7143	0.7606	0.0397	28.706	675.968	True
2	3961	122.826	46.3249	2.6514	0.9261	0.8431	0.4753	53.5506	452.253	False
3	547	38.4822	18.6369	2.0648	0.8749	0.9071	1.3303	28.7715	719.596	False
4	3031	74.7141	59.7338	1.2508	0.6007	0.7786	-0.0729	198.233	105.492	False
5	833	42.3386	25.9256	1.6331	0.7906	0.8976	0.935	290.168	253.631	False
6	8164	134.063	85.066	1.576	0.7729	0.8013	-0.8691	359.846	342.936	False
7	5658	108.902	72.88	1.4943	0.7431	0.7761	-0.0527	409.015	685.903	False
8	2786	87.8405	47.2353	1.8596	0.8431	0.8199	0.6054	469.408	431.352	False
9	4506	98.6377	61.8033	1.596	0.7794	0.8877	-0.0114	577.89	721.96	False
10	534	30.0222	23.0801	1.3008	0.6395	0.9287	0.8021	729.066	431.221	False
11	1086	59.0095	32.8062	1.7987	0.8312	0.7126	1.4627	783.261	210.357	False
12	4263	130.21	50.8324	2.5616	0.9207	0.6991	-1.4214	797.485	870.568	False
13	537	37.6344	28.0879	1.3399	0.6656	0.5582	-1.4625	836.14	208.279	False
14	6366	155.327	60.2608	2.5776	0.9217	0.7519	-1.2656	913.286	921.258	False
15	7862	142.695	80.4282	1.7742	0.826	0.8281	1.2173	982.477	693.681	True

Network per lacuna

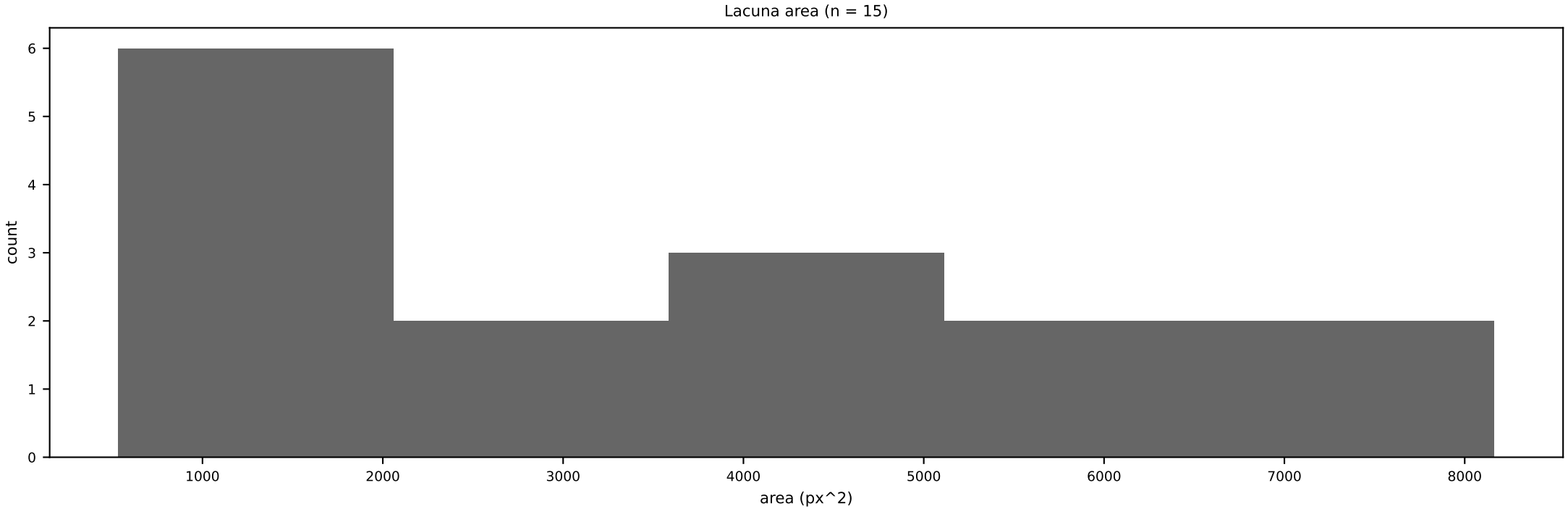
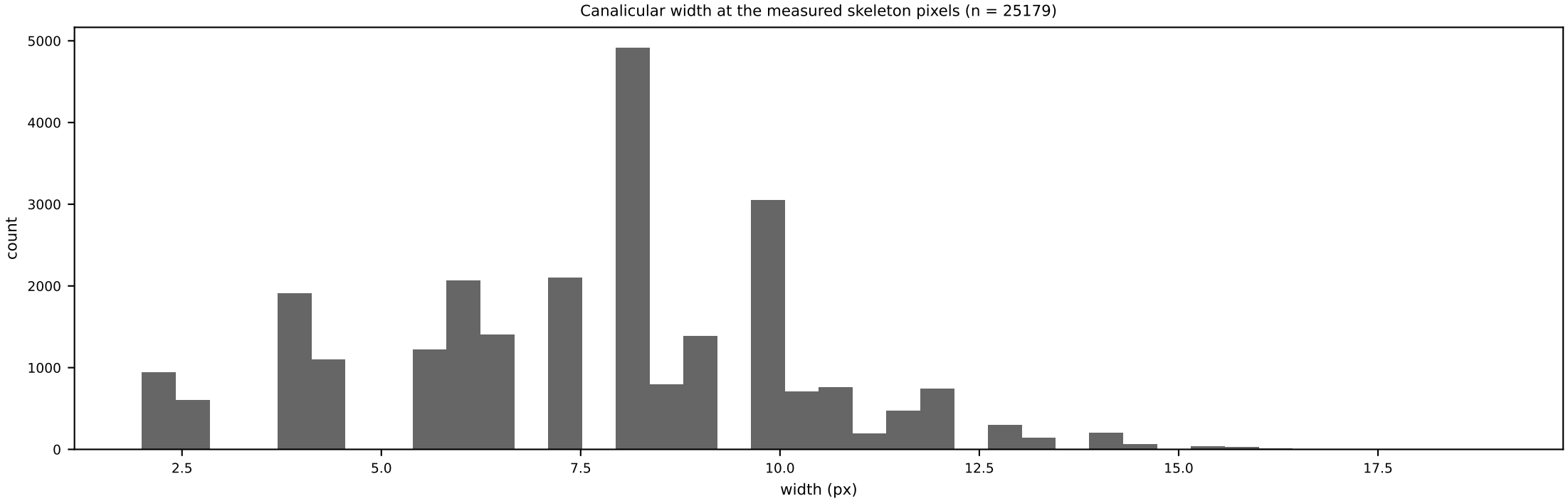
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	6	196	365	638.299	23	27.7521	6.7244
2	9	251	503	1112.97	68	16.3673	8.3083
3	6	177	234	243.823	16	15.239	6.0179
4	6	403	919	1063.92	49	21.7127	7.1285

5	8	231	523	1373.73	67	20.5034	8.4528
6	25	686	1221	2131.41	108	19.7353	7.7058
7	20	427	776	952.051	50	19.041	7.6251
8	6	373	723	1361.28	53	25.6846	8.1493
9	13	348	654	1562.55	72	21.7021	7.8628
10	2	99	384	17.0711	1	17.0711	6.0603
11	7	146	373	210.581	13	16.1985	6.9006
12	16	515	887	713.169	43	16.5853	6.9922
13	6	173	325	560.257	21	26.6789	5.5372
14	16	548	810	837.423	39	21.4724	8.1903
15	9	338	628	562.215	36	15.6171	7.3058

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	220.61	5180	11719	0.7389	0.6782	0.0378378	0.031146	2.7197
2	327.706	10056	22309	0.8579	0.7712	0.0249602	0.0225469	2.7464
3	102.326	3157	6458	0.9041	0.7441	0.0560659	0.0362341	5.8636
4	274.007	10301	25987	1	1	0.0391224	0.0353638	2.1897
5	121.983	5975	13793	1	1	0.0386611	0.0379178	6.5583
6	522.985	15465	29811	1	1	0.0443582	0.040958	4.7803
7	451.63	13578	29430	1	1	0.0314479	0.0263676	4.4284
8	268.777	10180	22542	1	1	0.0366405	0.0320735	2.2323
9	328.534	11289	25601	1	1	0.0308265	0.0255459	3.957
10	95.6396	5395	16431	1	1	0.0183503	0.0233705	2.0912
11	176.066	6367	15166	1	1	0.0229307	0.0245945	3.9758
12	441.789	13369	27297	1	1	0.0385219	0.0324944	3.6216
13	157.652	5421	12773	1	1	0.0319129	0.0254443	3.8059
14	478.902	14571	26738	0.9903	0.9107	0.037609	0.030294	3.341
15	502.902	11117	23341	0.7175	0.6712	0.0304039	0.0269055	1.7896

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	173	249	221.995	419.747	5	5	4
2	206	281	282.794	578.63	6	6	4
3	164	215	198.853	265.622	3	4	2
4	336	764	457.747	1041.06	8	14	14
5	199	447	278.735	609.227	6	7	6
6	663	997	787.311	1390.21	16	20	12
7	385	560	488.345	891.607	11	9	11
8	348	560	432.718	825.767	14	8	10
9	339	562	396.049	741.713	10	8	13
10	82	107	113.054	443.688	5	2	3
11	132	192	165.61	430.103	5	5	3
12	487	596	587.316	1010.31	12	12	6
13	133	176	194.752	368.605	4	4	1
14	521	686	619.973	918.536	15	13	16
15	309	422	388.019	704.813	8	5	7

548_hyp_HP_z08c1-2: distributions, pre-validation, px